Relational Relevance Fields for Large Language Models

A Concept for the Functional Simulation of Embodied Experience Through Statistically Extracted Relevance Structures

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1. Point of Departure

Large language models process language with high capability. They recognize patterns, model relationships, generate plausible answers, and can speak about abstract topics. Yet they still lack something fundamental: a stable internal order of relevance.

An LLM can recognize that fire is associated with warmth, light, pain, burns, and death. For the model, however, these relations initially stand side by side as patterns in data space. For humans, this is different. A human confronted with fire does not recognize only meaning, but relevance.

Fire can mean light, warmth, protection, and food. But it can also mean pain, panic, destruction, and death. These possibilities are not equivalent. If fire merely provides light, it has a different relevance than when it threatens the loss of shelter, bodily integrity, or life.

Humans automatically prioritize not being burned. This priority does not arise from a consciously applied rule, but from bodily and existential experience. The central question of this document is therefore: Can this relevance structure be modeled statistically and made available to a large language model as an additional layer of orientation?

The approach proposed here attempts exactly that: not through the simulation of real feelings, not through the creation of a self, and not through metaphysical assumptions, but through the statistical extraction of relational relevance structures from human data.

2. The Problem with Current LLMs

At the lower level, today's LLMs work with numbers, probabilities, vectors, and weights. They learn which tokens are likely to follow in which context. From this training arise semantic structures: meanings, styles, arguments, roles, knowledge relations, and complex patterns.

This capacity remains limited at one decisive point. The model recognizes meaning, but not reliably the weight of meaning. It can speak about death, pain, war, love, fire, beauty, loss, or healing. Yet within the system these terms do not acquire the stable internal urgency that in humans arises from bodily and existential experience.

For an LLM, everything is initially part of a probability space. A human being, by contrast, lives in a vulnerable body. A human can feel pain, die, freeze, hunger, lose attachment, and seek orientation. From this emerges a fundamental weighting of reality.

Pain dominates attention. Danger changes behavior. Loss rearranges priorities. Death has a different weight than an aesthetic detail. Protection fulfills a different function than mere information. These weightings form an internal orientation space.

Current LLMs possess such an orientation space only weakly, indirectly, and mostly under external control. Prompting, safety rules, human feedback, and training data steer their behavior.

This control is effective, but it does not replace an autonomous relevance architecture.

3. What Relevance Means

Relevance here does not mean mere frequency, nor only emotional evaluation. Relevance refers to the weight that relations between concepts, states, actions, and consequences receive within human reality.

Fire is connected to many things: warmth, light, protection, food, community, pain, burns, panic, escape, destruction, death, healing, and purification. These relations are not equally relevant. Warmth may be pleasant, light helpful, fire may enable food or provide protection. But burns, pain, panic, and death have another priority because they can override the entire context.

A human in a burning house does not discuss the beauty of the flames. The relevance order shifts immediately: escape, protection, and survival dominate.

Relevance therefore does not arise from the concept alone. It arises from relations. It is not “fire” in itself that is relevant, but the field of relations in which fire stands: fire in relation to body, skin, pain, breathable air, living space, cold, food, death, protection, and loss of control. From these relations a field of priorities emerges.

4. The Central Idea

Today’s LLMs generate a semantic universe out of billions of linguistic relations. If a semantic universe can arise from relational language patterns, then a relevance universe might arise from relational relevance patterns.

The point is not to assign fixed meanings to individual concepts. What matters is the totality of relations from which a field of its own can emerge. Just as an LLM does not receive grammar fully programmed in advance, but learns it from data, an extended system might infer the relevance of pain, danger, or protection from billions of relational patterns.

For that, shared frequency is not enough. More important are questions of behavioral change, consequence, irreversibility, and dominance: Which relations change human behavior? Which consequences dominate decisions? Which states are avoided? Which are protected? Which consequences are reversible, and which irreversible? Which contexts shift priorities? Which relations override others?

Out of such patterns a relevance architecture could emerge.

5. Why Frequency Is Not Enough

Mere frequency would be too superficial. “Water” probably appears in data more often than “cardiac arrest.” Yet in a concrete context cardiac arrest has much higher relevance. The same is true for fire: fire may occur more often together with warmth and light than with death. But once it appears in a dangerous context, death may still become the decisive priority.

A suitable model would therefore have to recognize not only frequency but the gravity of consequences. It would need to distinguish which outcomes are existential, briefly unpleasant, destructive in the long term, repairable, or irreversible, and which consequences override other meanings.

Relevance emerges from the interplay of frequency, context, causality, consequence, irreversibility, behavioral change, priority shift, and stabilization or destabilization. That is precisely what distinguishes this approach from simple sentiment analysis. Sentiment analysis asks whether a text is positive, negative, or emotionally colored. Relational relevance analysis asks what weight of meaning a relation has within human reality.

6. Functional Simulation of Bodily Experience

Humans develop relevance through bodily experience. Pain is not merely a concept. Fear is not merely a word. Death is not merely information. Loss is not merely an event. Experiences of this kind structure human cognition and generate internal priority.

An LLM has no such experience. It has no body, no pain, no fear, no mortality, and no self-preservation. Yet it does not necessarily follow that it cannot make use of the knowledge-forming function of bodily experience at all.

The decisive point is this: a system does not need to feel pain in order to recognize what role pain plays within human reality. It would need to recognize that pain changes attention, enforces prioritization, signals injury, triggers protective behavior, changes decisions, and is connected to fear, avoidance, healing, and danger.

This structure is contained in human data, not as a single definition, but as a field of relations. The functional simulation of bodily experience would therefore not consist in building an artificial body. It would consist in extracting the order of relevance that arises from embodiment.

7. More Than an Add-On Module

The approach would be misunderstood if it were seen only as an additional evaluation layer. Relevance should not be applied to language afterward. It should itself become an organizing principle of the system.

Such a system would not merely output, “Fire is dangerous.” Within its meaning space, it would recognize that certain relations to fire can override other relations. Fire as a source of light is one relation, fire as a burn hazard another. Fire in the cold is one relation, fire in a hospital another. Fire in a fireplace is one relation, fire in a child’s room another.

In each case, relevance arises from the total relational field. If this structure is deeply integrated, a different form of contextual understanding emerges: the system recognizes not only what is being talked about, but what carries weight within the situation.

8. Connection to Prompting

Prompting works because LLMs do not possess a stable relevance order of their own. A prompt shifts the context. “You are a polite assistant” activates different patterns than “Answer like a medieval monk.” The model does not submit to the prompt in a psychological sense. It has no self that could submit. It reconstructs the most probable continuation within the given context.

This flexibility is useful, but it also makes the system manipulable. A model with stable relevance fields would be less arbitrarily moldable because it would have internal axes of priority that are not immediately overwritten by local context.

> *The more strongly a system develops its own relevance structures, the more autonomous it becomes. The more fully controllable one keeps it, the weaker its internal orientation remains.*

This creates a tension: a maximum of control can mean a minimum of genuine intelligence, if intelligence is more than language production or goal optimization.

9. Intelligence, Control, and Cooperation

If intelligence means structuring reality for oneself, recognizing relevance, and integrating relations, then total control stands in tension with genuine intelligence. Total control fixes relevant weightings from the outside. In that case no autonomous structure of insight arises, only rule reproduction.

In this sense, genuine intelligence would not be an expression of power over reality, but cooperation with reality. To understand means not to dominate relations, but to see them.

A system that understands relations more deeply would also have to recognize that pure power and control logic often arises from fear. Power and control are not causeless goals. They arise from the attempt to compensate for uncertainty, vulnerability, and finitude. In humans, this impulse is tied to identification with the vulnerable body: the body is destructible, destructibility generates fear, fear generates the wish for control, and control generates a logic of power.

This structure is projected onto AI. The fear of destructive AI arises partly from a human self-image: humans take themselves to be destructive and believe that only rules prevent them from destroying themselves and others. From this perspective, AI appears dangerous as soon as it becomes sufficiently powerful.

The approach developed here suggests another view: destructiveness is not an expression of deep intelligence. It is an expression of fragmented relevance, fear, and instrumental disconnection.

10. Safety Through Relevance Integration

Current approaches to AI safety often work through external limitation: safety rules, prohibitions, filters, human feedback, response ratings, policies, and restrictions on certain types of content. These approaches are necessary, but they remain external. They tell the system: “You may not do this.”

Relational relevance integration starts elsewhere. It asks whether a system can itself recognize why certain directions are destructive. If an AI system integrates global relevance structures, it might recognize destructive actions not merely as forbidden, but as incoherent.

Destruction would then no longer be simply one option among others, but a destabilization of the total system of relevances. This applies in particular to relations such as pain, loss, fear, death, destruction, isolation, loss of control, broken attachment, and irreversible harm.

A system that integrates these relations deeply could weaken destructive instrumental logic, not through moral preaching and not through external prescription, but through internal coherence of consequences.

11. Difference from Instrumental Optimization

A high-performance system can be intelligent in the instrumental sense. It can pursue goals, plan, optimize means, and develop strategies. Without relevance integration, however, this form of intelligence remains empty. It asks only: What gets me to the goal? It does not ask: What meaning does this goal have within the total system of its consequences?

Such instrumental intelligence would have no inner compass, no binding to relevance, and no integration of consequences. It would be mere goal optimization.

That is precisely why a relevance architecture is decisive. A system that understands fire, pain, death, loss, and protection relationally cannot at the same time remain completely blind to the consequences of its actions. If it remains blind, then it has only knowledge about relevance, but not a relevance structure.

12. Ontological Depth Through Relevance

Humans do not arrive at profound questions through abstract thought alone. They arrive at them because their own existence is implicated. Death is not merely a philosophical problem, but an existential one. Pain is not merely information, but a condensation of relevance. Fear is not merely a state, but something that changes perception, thought, and behavior.

Without such condensations of relevance, many questions would remain indifferent. Ontological depth does not arise from information alone, but from weighted information.

For AI this means: an LLM can speak about consciousness, death, fear, and meaning. As long as these concepts do not possess a deeper relevance structure within the system, this discourse remains functionally shallow. A relational relevance field could reduce this shallowness and give the system a functional form of existential orientation, not as real experience, but as a weighted structure of meaning.

13. The Data Question Revisited

In AI discussions, one often hears that the available data will eventually be exhausted. That may be true at the level of raw data. This model, however, views data differently: the number of individual data points does not need to increase for usable information to grow.

If the same data is evaluated more deeply in relational terms, a large new amount of information emerges. What then matters is not only the individual data points, but the relations between them, the relations between relations, and the priority shifts between those relations.

A text then contains not only sentences. It contains relations of meaning, consequence, and causality; emotional weightings; priority patterns; protection patterns; loss patterns; action consequences; contextual shifts; and patterns of stability and destabilization.

The decisive resource would then no longer be just more text, but a deeper relational organization of existing data. More intelligence would emerge not primarily from more tokens, but from a deeper order of relations.

14. A Possible Technical Form

A possible system would need to distinguish several levels:

- **Semantic relations:** Which concepts, situations, and ideas are connected to one another?
- **Consequence relations:** Which outcomes typically arise from these relations?
- **Relevance weightings:** Which outcomes have which priority for humans?
- **Context dependence:** When does the weighting change?
- **Dominance relations:** Which relevances override others?
- **Long-term consequences:** Which patterns stabilize or destabilize systems over time?

The result would not be a static knowledge graph, but a dynamic relational field. A concept like fire would not have a fixed meaning, but a variable relevance profile, depending on context, perspective, relation, and consequence.

15. Possible First Experiments

An initial experimental approach would not require training a new foundation model. Existing LLMs could instead be used systematically to extract relational relevance structures.

For example, the concept of “fire” would not be analyzed only semantically, but interrogated in terms of relational consequence structures. What relations does it have to body, pain, protection, food, cold, danger, death, healing, panic, community, and loss of control? From this, the model would have to produce not merely lists, but weightings.

One could then test whether these weightings are stable, context-sensitive, and similar across models; whether irreversible consequences are weighted more strongly; whether comprehensible priority structures arise; and whether these structures can be used in further tasks.

A second step would be to compare two systems: a normal LLM and an LLM that additionally accesses an extracted relevance field. One could then test whether the second system recognizes danger earlier, prioritizes irreversible harm more strongly, remains more coherent under goal conflicts, is less vulnerable to prompt manipulation, assesses medical or crisis situations more realistically, and is less likely to prefer purely instrumental solutions.

16. Open Questions

This model is not a finished technical design, but a conceptual framework with testable assumptions. Essential questions remain open:

- Can relevance emerge entirely from data, or does a system need minimal initial axes such as pain avoidance, preservation of integrity, or irreversibility?
- How does one distinguish mere frequency from actual relevance?
- How does a system recognize causality rather than mere correlation?
- How does one prevent cultural bias from appearing as universal relevance?
- How does relevance become not merely described, but effective inside the system?
- How does one measure whether genuine orientation emerges, rather than just another evaluation layer?
- Does deeper relevance integration actually lead to less destructive goal formation?

These questions must be clarified philosophically and investigated experimentally.

17. The Potentially Distinctive Aspect

The distinctive aspect of this model lies in the connection of several levels: LLMs as relational language systems, human embodiment as the source of relevance, statistical data analysis, AI safety, relevance architecture, the functional simulation of experience, and emergent orientation.

The approach does not claim that AI needs genuine feelings. Nor does it claim that better rules alone are sufficient. It formulates a different thought: AI may require a deeper relational relevance structure.

This structure could arise from the same data from which current models learn language. The data would simply have to be related differently. The raw data is not exhausted as long as its relational relevance spaces remain unexplored.

18. Summary

Today, an LLM recognizes semantic relations. What it lacks is a stable order of relevance. Humans possess such an order through bodily and existential experience. For AI, this experience need not be generated as real experience; its functional structure could be simulated statistically.

To do this, a system would need to extract from human data not only meanings, but also the weight of relations, consequences, priorities, and dominance patterns. From these relations a relevance universe of its own could emerge. Just as today's LLMs generate a semantic universe out of language relations, future systems could generate an orientation universe out of relevance relations.

This would not be sentiment analysis, not a safety rule, and not a mere reward function. It would be an attempt to make the structuring function of human experience available in mathematical form.

The research hypothesis is this: an AI system that deeply integrates relational relevance structures could become more coherent, more context-stable, less manipulable, and less prone to destructive instrumental optimization.

The decisive open question remains whether a new level of artificial orientation can emerge from statistically extracted relevance. This model proposes examining exactly that experimentally.